Do R&D Tax Credits Promote High-Quality Innovation?

Evidence from US Patents, Citations, and Technology Proximity
around the introduction of the 1981 R&D Tax Credit.

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Abstract

This study evaluates the 1981 Federal R&D tax credit's impact on innovation quantity and quality. Using a dataset of 1.5 million patents (1975-1994) and a TWFE study, it compares US corporate responses against foreign ones. Results show a significant V-shapes recovery in patent volume after 1981. However, no board quality (measured by forward citations) emerged for US firms relative to foreign counterparts, suggesting that the policy has primarily increased marginal experimental development. Effects of the policy were heterogeneous: the volume of growth was driven by low exposure firms, while quality gains were concentrated among the top decile of established research leaders. Findings also confirm a four to ten year adjustment lag and stable technological drivers. The study concludes that the credit successfully expanded the innovator base but primarily increased output volume rather than shifting the technological frontier.

Keywords: R&D Tax Credit, Innovation Quality, Reaganomics, Event Study, Firm Heterogeneity.

I. Introduction

In January 1981, President Ronald Reagan assumed office and implemented a suite of neoliberal economic policies, known as "Reagonomics". His 1981 Program for Economic Recovery focused on four pillars: curtailing government spending growth, reducing the marginal tax rates on labor and capital, deregulation and controlling inflation. A centerpiece of his fiscal shift was the Economic Recovery Tax Act (ERTA), through which the Congress temporarily enacted a 25% tax credit available to drive research spending of businesses. While the credit was designed to increase research while protecting federal revenue, such a policy brings long-term effects beyond increased innovation. Subsequent legislation modified and extended the credit over time, including adjustments in the mid-1980s and later redesignations in the Internal Revenue Code. Together with the 1986 tax reform, these tax cuts, amongst the largest in U.S. history, are known as "Reagan tax cuts" of the "Reagan Revolution".

This research explores the efficiency of the 1981 federal R&D tax credit by addressing two key questions: (1) whether the credit increased the quality of innovation beyond its effect on patent counts; and (2) the temporal pattern of response among businesses after 1981.

II. Literature review

Current literature presents a dual perspective on the R&D tax credits impact on the scientific quality of innovations, often measured by forward citations. Analyzing the 1978 to 2010 period within the United States innovation and patent ecosystem, Melnik and Smyth demonstrate that on an aggregated level, state-level R&D tax credits seem to raise the private market value of patented research (by over 11%) and patent novelty without necessarily increasing the total quantity of patents, or their scientific quality (2024). These results also vary significantly based on the firm size; larger firms drive the market value and novelty without higher patent quantity, while smaller firms show a larger increase (approximately 20%) in patent quantity (Melnik and Smyth, 2024). For Initial Public Offering (IPO) firms, the presence and amount of R&D credits serve as a signal of innovation investment quality, which reduces information asymmetry and is positively associated with future patenting activity and forward citations (Hepfer et al., 2024). However, when looking at the effects in other countries, evidence from the United Kingdom (UK) suggests that R&D tax incentives can increase both R&D and quality-adjusted patenting, in particular for SMEs that are credit-constrained (Dechezleprêtre et al., 2016).

The responses to R&D tax incentives can be seen throughout time as adjustment frictions, rather than immediate effects. Analysis of the federal credit's first decade (1981–1991) in the US shows no discernible evidence that firms allocate or "retime" their research spending over time to maximize tax credits (Rao, 2016). A key concept is the user cost of R&D that drives firm decisions in matters of R&D spending; a 10% reduction of such cost leads to an immediate 11% increase in research intensity in the short run (Rao, 2016). Nonetheless, firms face significant adjustment costs as they only partially adjust spending initially and continue to increase it over

the longer run (Appelt et al., 2016; Rao, 2016). Indeed, major R&D projects typically have time horizons of three to ten years for phases like experimentation, testing, and commercial development (Brown, 1985). Early findings of modest effects were likely due to firms needing time to adapt their long-term planning to the credit's availability (Brown, 1985). Moreover, evidence shows that firms may learn and respond more to the incentives over time, as elasticity estimates for the federal credit were larger in the latter years of its first decade compared to the beginning (Rao, 2016).

The R&D tax credit significantly influences the direction of research and the diffusion of ideas across technology fields. This incentive can sometimes induce firms to change the types of technology they pursue, increasing patent novelty: the distance in technology class space between a firm's current and past portfolios (Melnik and Smyth, 2024). This can be partially explained by the fact that credits are significantly more effective at boosting experimental development (closer to market application) than basic or applied research, with the effect on development being more than three times as large (Appelt et al., 2023). The increase in novelty is most significant at narrower levels of technological classification (e.g. 4-digit CPC classes), suggesting firms are diversifying within their broader fields (Melnik and Smyth, 2024). Lastly, R&D induced by tax policies creates positive knowledge spillovers for technologically related firms, meaning the innovation of one firm leads to more innovation in neighboring technological areas (Dechezleprêtre et al., 2016). The total spillover effect on connected firms' patenting can be up to 1.7 times the direct effect on the firm receiving the credit (Dechezleprêtre et al., 2016).

III. Methodology

Not having access to firm-level financial data to measure the innovation input (expenditure), we measure innovation input/output with regards to patent citation or volume only. We implemented two complementary event-study designs around 1981.

The first compares US corporate assignees as treated group to foreign corporate assignees as control group, motivated by the fact that the federal research credit directly applies to U.S. taxpayers. It is thus assumed that the tax credit affects innovation activities within the US compared to their foreign counterparts. The incremental structure of the tax cut is important for interpretation: firms with higher baseline research activity or better capacity to scale R&D may be more likely to respond, and the incentive may differentially affect firms depending on their existing innovation intensity. This motivates our second “exposure” design that compares U.S. firms with higher versus lower pre-policy patenting intensity. This design reduces cross-country confounding, but exposure is proxied using historical patenting because firm-level R&D spending is not observed in the patent dataset.

These complementary designs allow us to observe whether the tax credit positive shock yields results on quantity or quality of innovation output, and to control whether these effects are linked to territorial effects or specialisation, ie. innovation intensity and US corporations.

A. Functional form

We model event-study difference-in-differences specifications around 1981 to examine dynamic changes in outcomes. The baseline event-study takes the form:

$$Y_{it} = \sum_{k \neq -1} \beta_k \times \mathbb{1}\{t - 1981 = k\} \times Treat_i + \varepsilon_{it} + \alpha_i + \gamma_t,$$

Where i indexes assignees, t indexes grand year, a_i assignees fixed effects, γ_t year fixed effects, ε_{it} error term, and β_k coefficients normalised for reference year $k =- 1$.

Patent counts are modeled in the code as $log(1 + patents)$ to avoid log(0). As citation outcomes are largely skewed towards zero and have long rightwing tails, we model with inverse hyperbolic function transformation $asinh(\bullet)$ that is loglike for large values and accommodates zeroes. Standard errors are clustered at the assignee level.

IV. Data collection

This paper uses the Kaggle "Patents and Patent Citations 1963–1999" dataset (Appendix 1), which contains (i) USPTO patent grant records and (ii) patent-to-patent citation links. Both files are matched by patent ID. Patent-level fields include grant year, assignee identifiers, country/state information, technology categories, and other attributes.

The patents file includes identifiers and metadata such as patent number (`PATENT`), grant year (`GYEAR`), application year (`APPYEAR`), assignee identifier (`ASSIGNEE`) and assignee-type code (`ASSCODE`), US technology class (`NCLASS`), and broad technology groupings (`CAT`, `SUBCAT`). The citation file provides directed edges from citing patents (`CITING`) to cited patents (`CITED`).

We read the patents file by selecting a standard set of columns needed for the analysis: `PATENT`, `GYEAR`, `APPYEAR`, `COUNTRY`, `POSTATE`, `ASSIGNEE`, `ASSCODE`, `NCLASS`, `CAT`, and `SUBCAT`. When available, we also load `GDATE` (a grant-date index used for finer-grained citation-window filtering). We coerce key identifier columns (`PATENT`, `GYEAR`, `APPYEAR`, `ASSIGNEE`, `ASSCODE`, and `GDATE` when present) to numeric values, using robust conversion rules that set non-parsable entries to missing values. Country/state and technology category variables are stored as string types.

Citations are loaded from `cite75_99.csv` using only `CITING` and `CITED`, coerced to numeric identifiers, and rows with missing IDs are dropped. This yields a clean citation edge list.

V. Data analysis

First, we have established an overview of the patenting landscape in the US from 1975 to 1994, which is the focal sample for this study. The research utilizes a dataset containing over 1.5 million focal patents and over 16.5 million citations. The chosen sample ends in 1994 to allow a 5-year forward citation window, ensuring that patents granted later than 1981 have time to accumulate citations. Second, we have narrowed down from the larger landscape into an isolation of US corporations and foreign corporations. More precisely, we explore whether US

corporations ("treated group" for the tax credit) behaved differently than foreign corporations ("control group") after 1981. The TWFE allows us to isolate the effect of the 1981 tax credit on both groups. Third, we assess whether and to which extent the pre-policy exposure to research activity of US corporations influenced their response to the 1981 tax credit. "High exposure" firms designate those in the top 25th percentile of patent production before 1981. This analysis tests whether the policy disproportionately benefited established innovators or encouraged the growth of smaller or less active firms. Lastly, we move beyond causal inference to predictive modeling, using a machine learning approach to identify the characteristics that define high-impact patents. By classifying patents in the top decile of five-year forward citations, this model evaluates how well pre-grant features can predict future technology quality.

A. Constructing needed variables and early exploratory data analysis

To measure forward citations using a fixed window without right-censoring, we define a focal cohort of patents granted between 1975 and 1994 inclusive. Because citations are observed through 1999, patents granted after 1994 would not have a full five-year citation window in the available data. The focal cohort is therefore restricted to ensure consistent observability of outcomes.

If `GDATE` is available, we compute a week-based lag as `lag_w = citing_gdate - cited_gdate` and keep only citations with `0 ≤ lag_w ≤ W`, where `W` approximates five years in weeks (with a small slack to account for rounding).
If `GDATE` is not available, we compute a year-based lag as `lag_y = citing_year - cited_year` and keep only citations with `0 ≤ lag_y ≤ 5`. This includes same-year citations at the year level.

For each focal patent *i*, we compute forward citations within five years (`fwd_cites_5y`) as the number of distinct citing patents within the window to avoid overweighting repeated citations from the same citing patent. We also compute self-citations within five years (`self_cites_5y`), i.e. forward citations where the citing and cited patents share the same assignee identifier (`ASSIGNEE`), again counted as distinct citing patents. Eventually, patents with no observed citations within the window are assigned zero forward citations and zero self-citations. Albeit present in the code, we ended up not using self-citations measures later on due to time constraints. A further development could be to distinguish external impact by excluding self-citations (i.e., using forward citations net of self-citations) and to analyze self-citations separately as potential strategic behavior.

These measures are visualised at the beginning of the code, as indicated by comments, to provide early exploratory visual data analysis before implementing the event-study.

B. Constructing Difference-in-Difference variables

We build an assignee-year panel from the focal patent-level dataset by grouping patents by assignee and grant year. For each assignee *a* in year *tt*, we compute:

- `patents_n`: number of patents granted to assignee aa in year tt,
- `fwd_cites_5y_sum`: total five-year forward citations across those patents,
- `mean_fwd_cites_5y = fwd_cites_5y_sum / patents_n`,
- analogous sums and means for self-citations[1]

The code leverages `ASSCODE` that describes conventional values of assignees. By dropping the first digit of the code with a euclidean division, we keep only the indication of the country of the inventor : 2 for US corporations as the treated group, 3 for foreign companies as control group. This treatment dummy (1 for US corporation, 0 otherwise) at the patent level is then matched to each corresponding assignee. For a given assignee, it qualifies it as treated (`treated_assignee`) if the average of all its patents is superior to a majority rule at 0.5.

Event time is defined relative to 1981 : $even_time = t - 1981$; with estimates for $k \in [-6, 10]$ and omitting $k = -1$ as the baseline or reference period. Event effects are then estimated for two outcomes on quality and quantity of patents.

VI. Results

A. Quantity and quality of granted patents after ERTA

Figure 1. Patent grants per year (1975-1993)

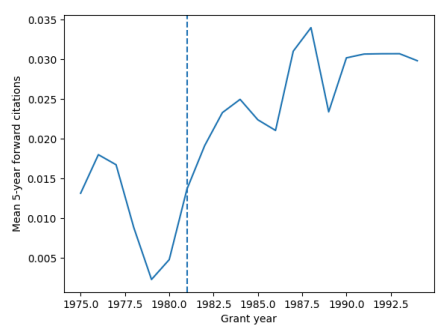

Figure 2. Mean forward citations per grant year (within 5 years)

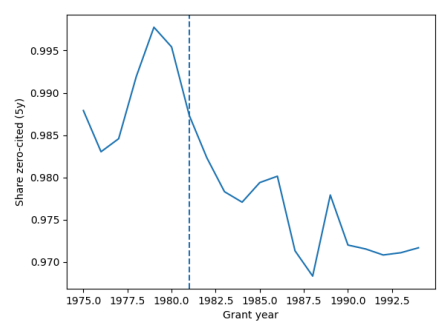

Figure 3. Share of patents with zero forward citations (within 5 years)

The pre-policy period (1975-1981) demonstrates a downward trend in patent granting, reaching a local minimum around 1979. Following the enactment of the tax credit in 1981, we observe a sustained upward trend in the number of patents granted annually. By 1994, the annual patent count exceeded 100,000, nearly doubling the number of patents in the late 1970s (Figure 1, Appendix 2).

[1] See remarks here above regarding the eventual restricted use of self-citations.

Going beyond patent counts, we observe that the mean 5-year forward citations began a sharp increase in 1980, rising approximately from 0.012 to 0.030 by 1980 (Figure 2). Subsequently, the share of patents with zero citations within the first five years dropped significantly after 1978, falling from nearly 99.8% in 1978 (local maximum) to approximately 97.2% in 1994 (Figure 3). These observations provide a first look at how ideas began to spread differently during that policy period, even if changes were noticeable even before 1981.

B. Impact on innovation quantity (US vs foreign corporations)

The model compares 94.477 corporate entities, using foreign corporations to account for global technological shocks. Coefficients before 1981, namely years -6 to -2, show a relatively stable patenting intensity between US and foreign firms, supporting the study's research design (Figure 4, Appendix 3). Post-1981 data reveals that while absolute patenting intensity rose for all groups, US corporations showed a relative decline in patenting intensity compared to the foreign corporations, with coefficients trending downward significantly through 1991. However, we observe that in Figures 4, 5, 6 and 7 the confidence intervals are wide and grow over time, likely due to a high variance in firm-level responses. Moreover, most post-period are statistically non-significant (Appendix 3).

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:          log_patents_n   R-squared:                       0.0318
Estimator:                   PanelOLS   R-squared (Between):            -0.0283
No. Observations:              218979   R-squared (Within):             -0.0924
Date:                Sat, Dec 20 2025   R-squared (Overall):            -0.0289
Time:                        17:02:54   Log-likelihood                -8.247e+04
Cov. Estimator:             Clustered
                                        F-statistic:                     255.26
Entities:                       94477   P-value                          0.0000
Avg Obs:                        2.3178   Distribution:               F(16,124470)
Min Obs:                        1.0000
Max Obs:                        17.000   F-statistic (robust):            2.7465
                                         P-value                          0.0002
Time periods:                       17   Distribution:               F(16,124470)
Avg Obs:                     1.288e+04
Min Obs:                        9325.0
Max Obs:                     1.747e+04
```

Table 1. OLS Estimation Summary (Patenting intensity)

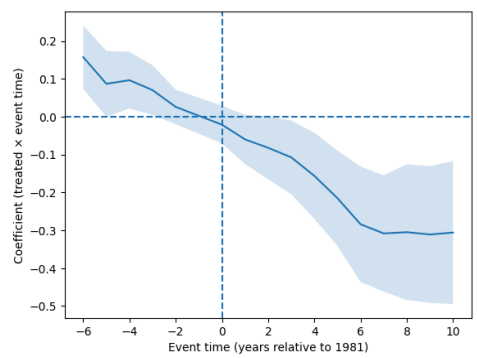

Figure 4. Event study: Patenting intensity (US corporations vs foreign corporations)

The event study for log-mean forward citations within five years shows that the 1981 credit did not yield a statistically significant relative increase in innovation quality for US corporations (Table 2).

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:     log_mean_fwd_cites_5y   R-squared:                       0.0003
Estimator:                       PanelOLS   R-squared (Between):             0.0204
No. Observations:                  218979   R-squared (Within):              0.0009
Date:                  Sat, Dec 20 2025   R-squared (Overall):             0.0103
Time:                          17:03:02   Log-likelihood                 4.012e+05
Cov. Estimator:                Clustered
                                          F-statistic:                      2.2822
Entities:                          94477   P-value                           0.0025
Avg Obs:                          2.3178   Distribution:               F(16,124470)
Min Obs:                          1.0000
Max Obs:                          17.000   F-statistic (robust):             0.5627
                                           P-value                           0.9133
Time periods:                         17   Distribution:               F(16,124470)
Avg Obs:                       1.288e+04
Min Obs:                          9325.0
Max Obs:                       1.747e+04
```

Table 2. OLS Estimation Summary (Citations intensity)

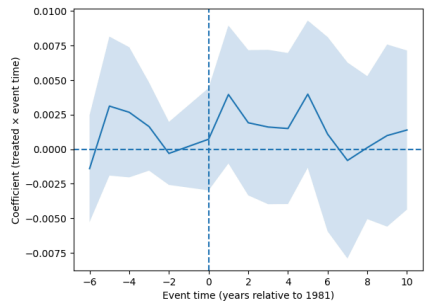

Figure 5. Event study: Citations intensity (US corporations vs foreign corporations)

Most post-1981 coefficients, for instance year 1 at 0.0040, have wide confidence intervals that cross the zero-line, indicating that the observed aggregate rise in quality (forward citations) was likely a global trend (Figure 5, Appendix 4). The lack of increased quality effect for US corporations, compared to foreign ones, suggests that the tax credit may have been more effective in stimulating the research volume (as seen in aggregate) than at shifting the quality frontier for domestic firms.

The divergence in patenting quantity between US and foreign firms was not immediate post-1981, the most significant relative changes appeared only four to ten years after the 1981 enactment. The negative relative trend in patent counts for US firms compared to foreign ones persisted through the end of the focal sample (1994), suggesting a long-term shift in the global competitive landscape rather than direct and temporary policy impact.

C. Pre-policy exposure to research activity and responses to 1981 tax credit

The event study for patenting quantity shows a dip for high exposure firms, relative to low exposure ones following the 1981 tax credit (Figure 6, Appendix 5). This relative decline suggests that firms with lower pre-existing research levels may have seen a higher proportional growth broadening the base of corporate innovators.

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:          log_patents_n   R-squared:                       0.0031
Estimator:                   PanelOLS   R-squared (Between):            -0.0029
No. Observations:               68787   R-squared (Within):              0.0463
Date:                  Sat, Dec 20 2025   R-squared (Overall):            -0.0026
Time:                          17:08:19   Log-likelihood               -2.498e+04
Cov. Estimator:              Clustered
                                          F-statistic:                      9.3330
Entities:                          20072   P-value                           0.0000
Avg Obs:                          3.4270   Distribution:                F(16,48683)
Min Obs:                          1.0000
Max Obs:                          17.000   F-statistic (robust):             17.478
                                           P-value                           0.0000
Time periods:                         17   Distribution:                F(16,48683)
Avg Obs:                          4046.3
Min Obs:                          2204.0
Max Obs:                          7249.0
```

Table 3. OLS Estimation Summary (Pre-policy exposure: Patenting)

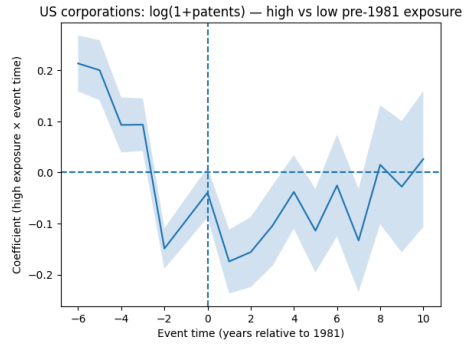

Figure 6. US corporations: High vs Low Pre-1981 Exposure (Patents)

In contrast to the volume of findings, the analysis of citation intensity reveals a positive and increasing trend for high-exposure firms. Starting around year 2 (1983), the coefficients for high-exposure firms begin to rise steadily (Figure 7, Appendix 6).

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:          log_patents_n   R-squared:                       0.0031
Estimator:                   PanelOLS   R-squared (Between):            -0.0029
No. Observations:               68787   R-squared (Within):              0.0463
Date:                Sat, Dec 20 2025   R-squared (Overall):            -0.0026
Time:                        17:08:19   Log-likelihood               -2.498e+04
Cov. Estimator:             Clustered
                                        F-statistic:                     9.3330
Entities:                       20072   P-value                          0.0000
Avg Obs:                       3.4270   Distribution:                F(16,48683)
Min Obs:                       1.0000
Max Obs:                       17.000   F-statistic (robust):            17.478
                                        P-value                          0.0000
Time periods:                      17   Distribution:                F(16,48683)
Avg Obs:                       4046.3
Min Obs:                       2204.0
Max Obs:                       7249.0
```

Table 4. OLS Estimation Summary (Pre-policy exposure: Citations)

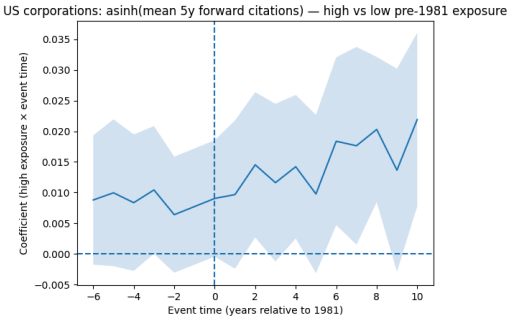

Figure 7. US corporations: High vs Low Pre-1981 Exposure (Citations)

Unlike the previous comparison of US and foreign corporations, several coefficients in the later periods of the quality study, such as years 6, 8 or 10, are statistically significant, as shown by parameter estimates (e.g. 0.0008 for Year 8, Appendix 7). Nonetheless, the model has a poor precision score for class 1, with among patents predicted as highly cited only 17.5% actually having this feature.

6. 4 Classifying the top decile of highly cited patents (predictive model)

The logistic regression model is trained on data through 1988 and tested on the 1989-1994 period. It provides a rigorous test of whether the drivers of innovation quality changed after the policy enactment.

```
Test AUC: 0.6658452136447872
              precision    recall  f1-score   support

           0      0.926     0.646     0.761     85061
           1      0.175     0.593     0.270     10774

    accuracy                          0.640     95835
   macro avg      0.551     0.620     0.516     95835
weighted avg      0.842     0.640     0.706     95835
```

Table 5. Test AUC Summary

The model achieved a Test AUC of 0.666, indicating a moderate but meaningful ability to distinguish high-impact patents from routine ones using only metadata such as country, assignee type and technology class. However, when contrasted with the precision of the model, this means that the model performs better than a random guess but lacks high predictive power. High-impact patents represent only 10% of the sample, thus the model used a balanced class weight to improve detection. The resulting recall of 0.593 for the high-cited class suggests that the model successfully identifies the majority of “superstar” innovators. The model maintains a high

precision for routine patents (0.926), ensuring that the features identified as "routine" are reliable.

VII. Discussion

This study presents a nuanced view of the effectiveness of the 1981 Federal R&D tax credit, aligned with the perspectives presented in current literature. The empirical data shows a V-shaped recovery in aggregate patent counts after 1981, yet the causal event study indicates that US firms did not significantly outperform foreign corporations in average citation quality. This aligns with Melink and Smyth's finding that tax credits often increase the private market value of research without necessarily raising its scientific quality or novelty. The 1981 credit seems to have stimulated a high volume of experimental development, meaning closer to market application innovation than fundamental research that drives citation levels. The results showcase a heterogeneity of response to R&D tax credits based on previous exposure to research. This result mirrors the theory that larger and more established firms drive novelty and market value, while smaller firms show a larger increase in patent quantity. This suggests that the credit served as a signal of investment quality for established research firms.

The event study results show that the most significant divergence in patenting occurred several years after 1981. This supports the theory of adjustment frictions, as noted by Brown, major R&D projects have time horizons of three to ten years for experimentation and development (1985). Moreover, firms likely required time to adapt long-term strategic planning to new fiscal incentives, which explains the relative shift in US patenting intensity that became stronger toward the late 1980s as firms responded more to incentives over time. While the credit lowered the user cost of R&D, the technological direction was still heavily influenced by a firm's existing niche.

VIII. Limitations

A significant limitation in this study is the concurrent shift in the US legal environment in the 1980s, more precisely the 1982 creation of the Court of Appeals for the Federal Circuit (CAFC). This could lead to a confounding variable that makes it difficult to isolate the tax credit's impact from the legal increased protection of patents. Moreover, using forward citations with a five-year window is a standard metric but not without flaws. This time span may have favored experimental development that gained instant market traction, and underrepresented the fundamental research that takes longer to be cited within the scientific community. While citations can serve as a signal of innovation quality, they do not capture the full private market value and novelty aspect within the given technology class. Lastly, relying on the ASSCODE possibly does not account for the complex R&D networks of multinational corporations that may have R&D offices in the US but are headquartered abroad.

Further limitations include weak statistical significance and pre-trend levels that do not satisfy the parallel trend hypothesis of DiD. Hence, these results are to be regarded as suggestive evidence and not causal results.

IX. Conclusion

This study concludes that the 1981 Federal R&D tax credit was an effective instrument for stimulating the quantity of innovation, reversing the downward pattern in patenting observed in the late 1970s. However, the effect on scientific quality was heterogeneous. While the credit did not empower the emergence of an increased quality among US innovations relative to foreign ones, it significantly enhanced the citation impact of patents produced by established and high-exposure US firms.

The temporal patterns observed confirm that R&D incentives do not provoke immediate responses, they require between three to ten years for firms to overcome adjustment frictions and align long-term research cycles with fiscal policy. Furthermore, the predictive model suggests that the fundamental drivers of innovation quality remain robust even in face of major fiscal shifts.

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The World in 1981—Short History—Department History—Office of the Historian. (n.d.)., from https://history.state.gov/departmenthistory/short-history/worldin1981

Appendix 1

```
Patents years: 1963 to 1999
Citations rows: 16522438
Has GDATE: True
Focal year range: 1975 to 1994
Focal patents rows: 1515262
Yearly tail:
    grant_year  patents_n  mean_fwd_cites_5y  median_fwd_cites_5y  \
10        1985      71661           0.022383                  0.0
11        1986      70860           0.021041                  0.0
12        1987      82952           0.031030                  0.0
13        1988      77924           0.033995                  0.0
14        1989      95537           0.023384                  0.0
15        1990      90364           0.030189                  0.0
16        1991      96513           0.030669                  0.0
17        1992      97444           0.030705                  0.0
18        1993      98342           0.030709                  0.0
19        1994     101676           0.029830                  0.0

    mean_self_cites_5y  share_zero_cites_5y
10            0.006949             0.979375
11            0.006336             0.980130
12            0.008788             0.971297
13            0.009278             0.968302
14            0.006636             0.977904
15            0.008919             0.971980
16            0.008652             0.971517
17            0.008169             0.970804
18            0.008796             0.971070
19            0.008389             0.971655
```

Appendix 2

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
evt_-6	0.1576	0.0426	3.7001	0.0002	0.0741	0.2410
evt_-5	0.0871	0.0442	1.9706	0.0488	0.0005	0.1738
evt_-4	0.0966	0.0381	2.5334	0.0113	0.0219	0.1714
evt_-3	0.0703	0.0334	2.1085	0.0350	0.0050	0.1357
evt_-2	0.0260	0.0232	1.1221	0.2618	-0.0194	0.0715
evt_0	-0.0208	0.0252	-0.8240	0.4099	-0.0703	0.0287
evt_1	-0.0598	0.0331	-1.8059	0.0709	-0.1246	0.0051
evt_2	-0.0819	0.0423	-1.9351	0.0530	-0.1649	0.0011
evt_3	-0.1072	0.0495	-2.1664	0.0303	-0.2042	-0.0102
evt_4	-0.1561	0.0580	-2.6929	0.0071	-0.2697	-0.0425
evt_5	-0.2151	0.0642	-3.3529	0.0008	-0.3409	-0.0894
evt_6	-0.2839	0.0777	-3.6555	0.0003	-0.4362	-0.1317
evt_7	-0.3081	0.0783	-3.9337	0.0001	-0.4616	-0.1546
evt_8	-0.3048	0.0914	-3.3358	0.0009	-0.4839	-0.1257
evt_9	-0.3109	0.0921	-3.3753	0.0007	-0.4914	-0.1304
evt_10	-0.3059	0.0962	-3.1803	0.0015	-0.4944	-0.1174

F-test for Poolability: 37.994
P-value: 0.0000
Distribution: F(94492,124470)

Appendix 3

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
evt_-6	-0.0014	0.0020	-0.7141	0.4751	-0.0053	0.0024
evt_-5	0.0031	0.0026	1.2165	0.2238	-0.0019	0.0081
evt_-4	0.0027	0.0024	1.1117	0.2663	-0.0020	0.0074
evt_-3	0.0016	0.0016	1.0063	0.3143	-0.0015	0.0048
evt_-2	-0.0003	0.0012	-0.2642	0.7916	-0.0026	0.0020
evt_0	0.0007	0.0019	0.3818	0.7026	-0.0030	0.0044
evt_1	0.0040	0.0025	1.5560	0.1197	-0.0010	0.0089
evt_2	0.0019	0.0027	0.7139	0.4753	-0.0033	0.0072
evt_3	0.0016	0.0028	0.5633	0.5732	-0.0040	0.0072
evt_4	0.0015	0.0028	0.5365	0.5916	-0.0040	0.0069
evt_5	0.0040	0.0027	1.4668	0.1424	-0.0013	0.0093
evt_6	0.0011	0.0036	0.3033	0.7616	-0.0059	0.0081
evt_7	-0.0008	0.0036	-0.2261	0.8211	-0.0079	0.0063
evt_8	0.0001	0.0026	0.0432	0.9656	-0.0050	0.0053
evt_9	0.0010	0.0034	0.2934	0.7692	-0.0056	0.0076
evt_10	0.0014	0.0029	0.4728	0.6363	-0.0044	0.0071

F-test for Poolability: 0.8832
P-value: 1.0000
Distribution: F(94492,124470)

Appendix 4

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
evt_-6	-0.0014	0.0020	-0.7141	0.4751	-0.0053	0.0024
evt_-5	0.0031	0.0026	1.2165	0.2238	-0.0019	0.0081
evt_-4	0.0027	0.0024	1.1117	0.2663	-0.0020	0.0074
evt_-3	0.0016	0.0016	1.0063	0.3143	-0.0015	0.0048
evt_-2	-0.0003	0.0012	-0.2642	0.7916	-0.0026	0.0020
evt_0	0.0007	0.0019	0.3818	0.7026	-0.0030	0.0044
evt_1	0.0040	0.0025	1.5560	0.1197	-0.0010	0.0089
evt_2	0.0019	0.0027	0.7139	0.4753	-0.0033	0.0072
evt_3	0.0016	0.0028	0.5633	0.5732	-0.0040	0.0072
evt_4	0.0015	0.0028	0.5365	0.5916	-0.0040	0.0069
evt_5	0.0040	0.0027	1.4668	0.1424	-0.0013	0.0093
evt_6	0.0011	0.0036	0.3033	0.7616	-0.0059	0.0081
evt_7	-0.0008	0.0036	-0.2261	0.8211	-0.0079	0.0063
evt_8	0.0001	0.0026	0.0432	0.9656	-0.0050	0.0053
evt_9	0.0010	0.0034	0.2934	0.7692	-0.0056	0.0076
evt_10	0.0014	0.0029	0.4728	0.6363	-0.0044	0.0071

F-test for Poolability: 0.8832
P-value: 1.0000
Distribution: F(94492,124470)

Appendix 5

PanelOLS Estimation Summary

Dep. Variable:	log_patents_n	R-squared:	0.0031
Estimator:	PanelOLS	R-squared (Between):	-0.0029
No. Observations:	68787	R-squared (Within):	0.0463
Date:	Sat, Dec 20 2025	R-squared (Overall):	-0.0026
Time:	17:08:19	Log-likelihood	-2.498e+04
Cov. Estimator:	Clustered		
		F-statistic:	9.3330
Entities:	20072	P-value	0.0000
Avg Obs:	3.4270	Distribution:	F(16,48683)
Min Obs:	1.0000		
Max Obs:	17.000	F-statistic (robust):	17.478
		P-value	0.0000
Time periods:	17	Distribution:	F(16,48683)
Avg Obs:	4046.3		
Min Obs:	2204.0		
Max Obs:	7249.0		

Appendix 6

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
evt_-6	0.0088	0.0054	1.6344	0.1022	-0.0017	0.0193
evt_-5	0.0100	0.0061	1.6321	0.1027	-0.0020	0.0219
evt_-4	0.0083	0.0057	1.4718	0.1411	-0.0028	0.0195
evt_-3	0.0104	0.0053	1.9658	0.0493	3.064e-05	0.0208
evt_-2	0.0064	0.0048	1.3251	0.1852	-0.0031	0.0158
evt_0	0.0090	0.0048	1.8670	0.0619	-0.0005	0.0185
evt_1	0.0097	0.0062	1.5693	0.1166	-0.0024	0.0217
evt_2	0.0145	0.0060	2.4062	0.0161	0.0027	0.0264
evt_3	0.0116	0.0066	1.7709	0.0766	-0.0012	0.0245
evt_4	0.0142	0.0060	2.3817	0.0172	0.0025	0.0259
evt_5	0.0098	0.0066	1.4861	0.1373	-0.0031	0.0226
evt_6	0.0184	0.0070	2.6332	0.0085	0.0047	0.0320
evt_7	0.0176	0.0082	2.1448	0.0320	0.0015	0.0337
evt_8	0.0203	0.0060	3.3648	0.0008	0.0085	0.0321
evt_9	0.0136	0.0084	1.6181	0.1057	-0.0029	0.0302
evt_10	0.0219	0.0072	3.0375	0.0024	0.0078	0.0360

F-test for Poolability: 0.7329
P-value: 1.0000
Distribution: F(20087,48683)

Included effects: Entity, Time